

Policy Memo: Why Limiting Climate Change Matters for Niger

To: Minister of Environment, Head of Niger's UNFCCC delegation	From: KOUAME Koffi Fidèle, Climate Adviser
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Key message

Models indicate that, under current global policies, Niger could face about 91 days of dangerous heat a year by 2100, three times today's level, against about 42 if warming is held below 2 °C and 32 on a 1.5 °C pathway. Heat is where faster global emission cuts make the clearest difference: even the low end of the current-policies range is above the central below-2 °C value. Severe drought, by contrast, is projected to affect most of the country in every scenario, so Niger also needs adaptation finance now.

Three priority impacts

Impact (CIE indicator)	Today (2020)	2100: 1.5 °C path	2100: below 2 °C	2100: current policies
Dangerous heat: days per year with a daily maximum heat index above 40 °C	31 [25–38]	32 [24–50]	42 [29–72]	91 [59–134]
Severe drought: share of land with at least one month of SPEI below –1.5	84 % [72–97]	87 % [66–100]	98 % [81–100]	100 % [98–100]
Extreme rainfall: annual maximum 5-day rainfall, change from 1995–2014	+8 % [–7, +29]	+9 % [–6, +38]	+17 % [–1, +54]	+43 % [+5, +110]

Table 1. Source: Climate Impact Explorer (Climate Analytics, 2026), Niger, annual values, area-weighted average. NGFS scenarios: Net Zero 2050 (1.3 °C of global warming in 2100), Below 2 °C (1.6 °C) and Current policies (2.9 °C), as given in the CIE data. Median of five climate models (18 for extreme rainfall); 5–95 % range in brackets. Values after 2060 are indicative.

Dangerous heat. The number of days with a daily maximum heat index above 40 °C, the “danger” threshold used in heat warnings, is projected to rise from about 31 today to 91 by 2100 under current policies, against 42 below 2 °C. The two pathways only separate after 2030 (Figure 1). This matters because agriculture employs over 80 % of the population (World Bank, 2024), and farming and herding are outdoor work.

Severe drought. This indicator accounts for evaporation as well as rainfall. About 84 % of the land already experiences at least one month of severe drought a year, and models project 98 to 100 % by 2100 both below 2 °C and under current policies. The 1.5 °C pathway has a lower central value (87 %), but its range (66–100 %) largely overlaps that of the below-2 °C scenario (81–100 %), so the models do not clearly separate the two. Rain-fed millet, sorghum and pastures are therefore likely to face more frequent water stress whatever the global outcome.

Extreme rainfall. The heaviest five-day rainfall of the year could intensify by about 43 % by 2100 under current policies, against 17 % below 2 °C, although the model range is wide. Heavier downpours would raise flood risk in Niamey and the Niger River valley, where people, infrastructure and irrigated rice are concentrated.

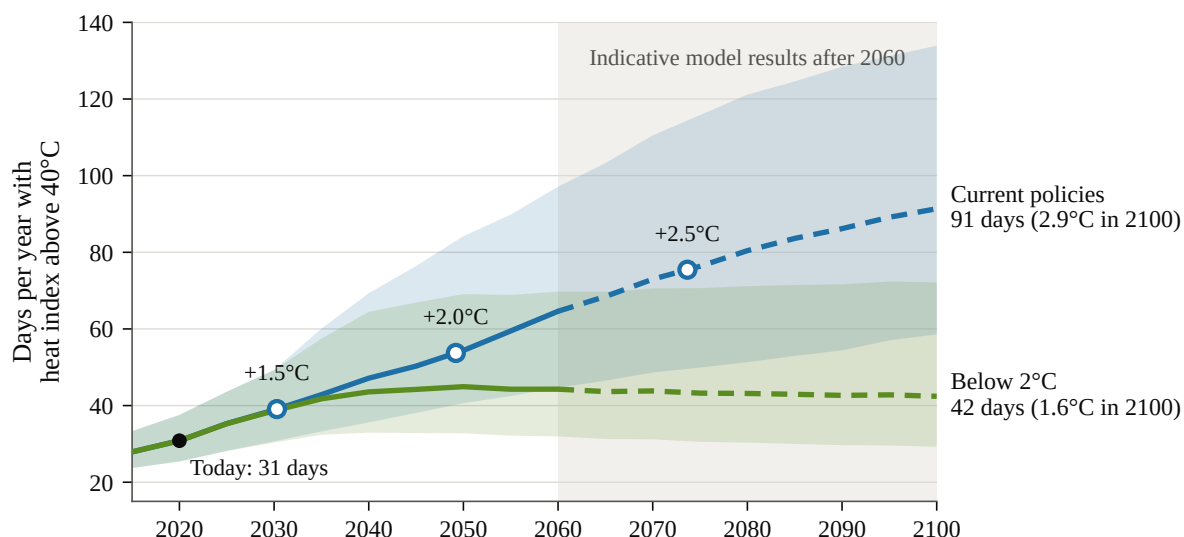


Figure 1. Days of dangerous heat per year in Niger, below 2 °C and under current policies. The curves are identical until about 2030; the gap that opens afterwards is what limiting warming avoids. Open circles mark the years in which global warming reaches +1.5 °C (about 2030), +2.0 °C (about 2049) and +2.5 °C (about 2074) on the current-policies pathway. Shaded bands show the 5–95 % model range; dashed lines and the grey area show indicative results after 2060. Chart drawn from Climate Impact Explorer data (Climate Analytics, 2026).

Implications for Niger’s national interest

Limiting warming is in Niger’s direct interest. Compared with current policies, a below-2 °C world would avoid about 49 days of dangerous heat a year by 2100 and reduce the projected intensification of extreme rainfall from 43 % to 17 %; a 1.5 °C pathway would keep heat close to today’s level. The World Bank estimates that climate change could reduce Niger’s annual GDP by 2.2 % in wetter scenarios and by 11.9 % in drier ones by 2050 (World Bank, 2022). Given the country’s dependence on rain-fed farming and herding, these losses are likely to cascade: a hot, dry year cuts harvests and pasture, raises grain prices, forces households to sell livestock, and can deepen malnutrition and tensions between farmers and herders. Because drought is projected to spread in every scenario, Niger should defend the 1.5 °C goal in the negotiations while seeking adaptation finance for drought-tolerant crops, water harvesting and early warning.

Uncertainty

These figures are projections, not forecasts, and model ranges are wide (59 to 134 heat days in 2100 under current policies). The heat index relies on an estimate of humidity, and the drought indicator counts a single month of drought, so it saturates. Given Niger’s high vulnerability, actual impacts may exceed what these hazard indicators suggest.

References

- Climate Analytics (2026). *Climate Impact Explorer: Niger*. Indicators: days per year with dangerous heat risk ($HI > 40^{\circ}\text{C}$), area under severe drought ($SPEI < -1.5$) and annual maximum 5-day precipitation; NGFS Net Zero 2050, Below 2°C and Current Policies scenarios; ISIMIP data. <https://climate-impact-explorer.climateanalytics.org> (accessed 9 October 2026).
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- World Bank (2024, 28 June). The World Bank supports food security and climate resilience for households in Niger [Press release]. <https://www.worldbank.org/en/news/press-release/2024/07/01/the-world-bank-supports-food-security-and-climate-resilience-for-households-in-niger>